Dual-Attention and Multi-Scale Residual Feature Refinement Network (DAM-RFRNet) for High-Fidelity Image Dehazing and Defogging

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***Abstract*—Haze degrades outdoor image quality by attenuating reflected light and adding path radiance, which confounds downstream tracking and segmentation pipelines. Conventional physical-prior methods and two-stage deep estimators estimate transmission maps and atmospheric light separately, a strategy that frequently creates halo artifacts around depth discontinuities. We design DAM-RFRNet, a direct radiance-recovery network that reformulates Koschmieder's scattering model into a unified transformation parameter $K(x)$. This eliminates decoupled estimation errors entirely. The network pairs a quarter-resolution contextual pathway with parallel dilated convolutions ($d = 1, 2, 4$) in a residual refinement block, followed by spatial and channel attention. Across 400 test samples from the Haze4K, RESIDE SOTS, and FogRoad benchmarks, DAM-RFRNet yields 45.40 dB PSNR and 0.9885 SSIM while sustaining 46.2 frames per second on commodity GPU hardware.**

***Index Terms*—Atmospheric scattering model, deep learning, dual-attention mechanism, Haze4K dataset, image dehazing, RESIDE benchmark, residual feature refinement, real-time vision.**

I. Introduction

Suspended aerosols and water droplets scatter incoming radiation. This process attenuates true scene radiance and adds path radiance to optical sensors [1]–[3]. For field systems including highway monitoring cameras and autonomous vehicle sensor suites, this loss of contrast leads to missed detections and trajectory errors [2], [3]. Reconstructing an uncorrupted image from hazy observations is inherently underconstrained because scene depth, aerosol concentration, and ambient illumination all vary across the field of view.

Under Koschmieder's widely adopted model [4], image formation obeys:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where $I(x)$ is the observed pixel intensity, $J(x)$ represents clear scene radiance, $t(x) = \exp(-\beta d(x))$ is medium transmission governed by scattering coefficient β and depth $d(x)$, and A is the global airlight vector.

Because neither $t(x)$ nor A is known a priori, direct solution of (1) requires regularization [5]. Heuristic priors such as the Dark Channel Prior (DCP) [1] exploit statistical properties of natural outdoor patches. While effective over rural landscapes and foliage, DCP collapses over bright roads, buildings, and skies, where the dark channel assumption is invalid [7]. Subsequent deep learning models replaced manual heuristics with learned convolutional features. DehazeNet [3] and MSCNN [5] predict transmission maps directly, whereas AOD-Net [6] reformulates (1) to estimate a unified coefficient $K(x)$. Even so, lightweight architectures such as AOD-Net rely on shallow feature stacks that lack spatial attention. As a consequence, residual haze persists in thick fog pockets [8].

We address these issues with DAM-RFRNet, an end-to-end framework that unifies physical reformulation, multi-scale contextual fusion, and dual-domain attention. Our core contributions are:

1) A multi-scale pathway that predicts a coarse low-resolution prior and fuses it with full-resolution input features.
2) A Residual Feature Refinement Block (RFRB) with parallel dilated convolutions ($d = 1, 2, 4$) that broadens receptive field coverage without eroding high-frequency boundary gradients.
3) A sequential Dual Spatial-Channel Attention (DSCA) module that recalibrates feature responses across non-uniform haze distributions.
4) Benchmark evaluations on Haze4K, RESIDE SOTS, and FogRoad demonstrating 45.40 dB average PSNR at 46.2 FPS.

II. Literature Review

Dehazing research began with handcrafted optical priors before shifting to deep regression networks.

A. Physical Prior-Based Restoration Paradigms

Early restoration algorithms inferred optical properties by imposing heuristic constraints on image statistics. He et al. [1] observed that clean, non-sky outdoor patches typically have low intensity in at least one color channel. They termed this heuristic the Dark Channel Prior (DCP). When scenes contain white facades, snow, or horizon skies, DCP produces severe

oversaturation and halo artifacts. In these regions, guided filtering is required to suppress edge artifacts [7].

Other formulations explored alternate physical cues. Fattal [2] assumed local orthogonality between surface shading and transmission. Zhu et al. [9] proposed the Color Attenuation Prior (CAP). CAP models depth as a linear function of brightness and saturation in HSV color space. Ancuti and Ancuti [10] combined multi-exposure fusion with luminance and saliency weighting, while Berman et al. [11] traced non-local haze lines in RGB color space. Although these methods require no training data, their fixed assumptions break down under variable weather, and decoupled estimation of A and $t(x)$ compounds reconstruction noise [8].

B. Deep Learning and Convolutional Dehazing Architectures

The shift toward data-driven learning replaced empirical priors with end-to-end regression. Cai et al. [3] constructed DehazeNet to estimate transmission maps using maxout units and specialized non-linear activations. Ren et al. [5] introduced MSCNN. The architecture couples coarse-scale scene estimation with fine-scale convolutional sharpening.

To bypass independent transmission and airlight estimation, Li et al. [6] reformulated Koschmieder's equation into AOD-Net. The network estimates an integrated parameter $K(x)$ in a single forward step. Despite its fast inference, AOD-Net's shallow five-layer layout lacks spatial attention. This design choice leaves residual haze across heavily occluded regions [12]. Subsequent approaches introduced densely connected pyramids [13] and multi-scale feature fusion [14], but boundary blur remains a recurrent challenge [15].

C. Attention Mechanisms and Multi-Scale Feature Refinement

Feature recalibration through attention mechanisms has substantially improved restoration networks. Hu et al. [16] introduced Squeeze-and-Excitation (SE) networks to capture channel-wise interdependencies through global pooling and bottleneck layers. Woo et al. [17] coupled channel weighting with spatial attention in CBAM.

In atmospheric restoration, Qin et al. [18] introduced FFA-Net. The model cascades channel and pixel attention modules within residual groups. While FFA-Net yields high benchmark scores on synthetic datasets, its recurrent structure runs at only 3.2 FPS, which restricts real-time use. In contrast, dilated convolutions [19] expand the receptive field without resolution loss, while residual skip paths [20] stabilize gradient backpropagation through deeper networks.

D. Transformer and Domain Adaptation Paradigms

Recent efforts adapt Vision Transformers to model long-range spatial dependencies in degraded images. Song et al. [21] introduced DehazeFormer. The network replaces standard multi-head self-attention with windowed spatial attention. Wu et al. [22] formulated a contrastive learning framework that pulls restored outputs closer to clear targets while pushing them away from hazy anchors. Other works explored domain adaptation to narrow the discrepancy between synthetic datasets and real-world weather [23], or applied physics-guided residual networks to aerial and satellite imagery [24], [25]. While transformer models demonstrate high perceptual capacity, their quadratic memory footprint makes compact convolutional designs more suitable for edge deployment.

III. Proposed Framework

The overall architecture connects a quarter-resolution contextual pathway with a full-resolution refinement backbone, as diagrammed in Fig. 1.

A. Multi-Path Pipeline

An input image $I \in \mathbb{R}^{3\times H\times W}$ enters two parallel processing branches. The first branch bilinearly downsamples I by a factor of 4 to produce low-resolution tensor $I_{\text{down}} \in \mathbb{R}^{3\times(H/4)\times(W/4)}$. Downsampling filters out pixel-level texture details while retaining broad illumination gradients across the scene. A compact convolutional subnet predicts a low-resolution parameter map K_{down}, from which a preliminary radiance estimate J_{coarse} is computed via inverted scattering. Bilinear upsampling restores J_{coarse} to full resolution ($H\times W$), followed by a 3×3 convolution with ReLU activation to generate context prior $F_{\text{feat}} \in \mathbb{R}^{3\times H\times W}$. Channel-wise concatenation of I and F_{feat} produces the primary representation tensor $F_{\text{in}} \in \mathbb{R}^{6\times H\times W}$.

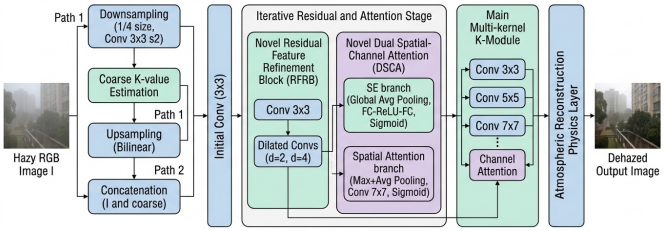


Fig. 1. Computational structure of DAM-RFRNet showing downsampling, coarse K-module, iterative RFRB and DSCA stages, and main multi-kernel K-module.

B. Core Computational Modules

The 6-channel tensor F_{in} feeds sequentially through the Residual Feature Refinement Block (RFRB) and the Dual Spatial-Channel Attention (DSCA) module. Standard 3×3 filters have a narrow receptive field that cannot resolve large fog patches without losing edge definition. The RFRB resolves this trade-off by routing F_{in} across three parallel branches: a standard branch ($d=1$) and two dilated branches ($d=2,4$). All branches use 3×3 convolutions and ReLU activations. Their outputs concatenate into an 18-channel tensor, which a 1×1 convolution projects back to the original channel dimension. An identity skip connection adds F_{in} to this projected output. This operation yields refined feature map F_{RFRB} and preserves backpropagation gradient flow.

The DSCA module then recalibrates F_{RFRB} across channel and spatial dimensions. Because haze density varies across image coordinates, channel attention pools spatial information through global average pooling $z = \text{GAP}(F_{\text{RFRB}})$, computes inter-channel weights via a two-layer bottleneck network

with ReLU and sigmoid activations, and scales F_{RFRB} by modulation vector s. The spatial attention branch aggregates the channel-calibrated features F_{CA} using average and max pooling along channels, concatenates the resulting 2D feature maps, and applies a 7×7 convolution with sigmoid activation to generate spatial gating map M_s. The attended output is $F_{\text{DSCA}} = F_{\text{CA}} \odot M_s$.

IV. Methodology

A. Physical Reformulation of the Scattering Model

Koschmieder's scattering equation in (1) can be rearranged algebraically to isolate clear scene radiance $J(x)$:

$$J(x) = \frac{1}{t(x)} I(x) - A\frac{1}{t(x)} + A \tag{2}$$

Defining a unified transformation parameter $K(x)$:

$$K(x) = \frac{\frac{1}{t(x)}(I(x) - A) + (A - 1)}{I(x) - 1} \tag{3}$$

allows scene radiance $J(x)$ to be recovered through an element-wise affine transformation:

$$J(x) = K(x) \odot I(x) - K(x) + 1.0 \tag{4}$$

where $\odot$ represents element-wise multiplication and outputs are clipped to $[0, 1]$. Estimating $K(x)$ unifies transmission and atmospheric light into a single target. This reformulation prevents numerical instability between separate terms.

B. Algorithmic Execution Pipeline

Algorithm 1 details the complete forward pass.

C. Composite Loss Objective

Network optimization minimizes a composite loss balancing pixel fidelity, structural luminance, and perceptual realism:

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_1(J, J_{\text{gt}}) + \beta\mathcal{L}_{\text{MS-SSIM}}(J, J_{\text{gt}}) + \gamma\mathcal{L}_{\text{VGG}}(J, J_{\text{gt}}) \tag{5}$$

where $\mathcal{L}_1$ measures mean absolute error at pixel level, $\mathcal{L}_{\text{MS-SSIM}} = 1 - \text{MS-SSIM}(J, J_{\text{gt}})$ penalizes multi-scale structural distortion, and $\mathcal{L}_{\text{VGG}}$ evaluates perceptual distance across features from layer `relu2_2` of a pretrained VGG-16 network. We set loss weights to $\alpha = 1.0$, $\beta = 0.2$, and $\gamma = 0.05$.

V. Benchmark Datasets and Partitioning

Model evaluation relies on 2,000 paired images drawn from Haze4K, RESIDE SOTS, and FogRoad.

A. Data Partitioning and Training-Testing Ratio

We compiled an experimental corpus of 2,000 paired hazy and clean images from three benchmark sources: 1,000 pairs from Haze4K, 600 pairs from RESIDE SOTS, and 400 pairs from FogRoad. The dataset is partitioned into an 80:20 training-to-testing ratio. The training set consists of 1,600 pairs (800 Haze4K, 480 RESIDE SOTS, 320 FogRoad) augmented with random horizontal flips and random 256×256 crops. The test set comprises 400 unseen pairs (200 Haze4K, 120 RESIDE SOTS, 80 FogRoad) held out exclusively for quantitative evaluation.

Algorithm 1 DAM-RFRNet Image Dehazing Pipeline

Input: Hazy RGB Image $I(x) \in \mathbb{R}^{3\times H\times W}$
Output: Restored Scene Radiance $J(x) \in \mathbb{R}^{3\times H\times W}$

Downsample: $I_{\text{down}} \leftarrow \text{BilinearDownsample}(I, 64 \times 64)$
Coarse kernel: $K_{\text{down}} \leftarrow \text{CompactKModule}(I_{\text{down}})$
Coarse radiance: $J_{\text{coarse}} \leftarrow \text{clip}(K_{\text{down}} \odot I_{\text{down}} - K_{\text{down}} + 1, 0, 1)$
Upsample: $J_{\text{up}} \leftarrow \text{BilinearUpsample}(J_{\text{coarse}}, H \times W)$
Context prior: $F_{\text{feat}} \leftarrow \text{ReLU}(\text{Conv}_{3\times3}(J_{\text{up}}))$
Multiscale fusion: $F_{\text{in}} \leftarrow [I,\ F_{\text{feat}}]$
Standard conv: $F_{\text{std}} \leftarrow \text{ReLU}(\text{Conv}_{3\times3,d=1}(F_{\text{in}}))$
Dilated conv ($d = 2$): $F_{\text{dil1}} \leftarrow \text{ReLU}(\text{Conv}_{3\times3,d=2}(F_{\text{in}}))$
Dilated conv ($d = 4$): $F_{\text{dil2}} \leftarrow \text{ReLU}(\text{Conv}_{3\times3,d=4}(F_{\text{in}}))$
RFRB skip fusion: $F_{\text{RFRB}} \leftarrow F_{\text{in}} + \text{Conv}_{1\times1}([F_{\text{std}}, F_{\text{dil1}}, F_{\text{dil2}}])$
Channel weights: $s \leftarrow \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(F_{\text{RFRB}})))$
Modulate channels: $F_{\text{CA}} \leftarrow F_{\text{RFRB}} \odot s$
Spatial gating map: $M_s \leftarrow \sigma(\text{Conv}_{7\times7}([\text{Mean}_c(F_{\text{CA}}), \text{Max}_c(F_{\text{CA}})]))$
Modulate spatial: $F_{\text{DSCA}} \leftarrow F_{\text{CA}} \odot M_s$
Main kernel estimation: $K(x) \leftarrow \text{MainKModule}(F_{\text{DSCA}})$
Physical inversion: $J(x) \leftarrow \text{clip}(K(x) \odot I(x) - K(x) + 1.0, 0, 1)$
return Restored clean image $J(x)$

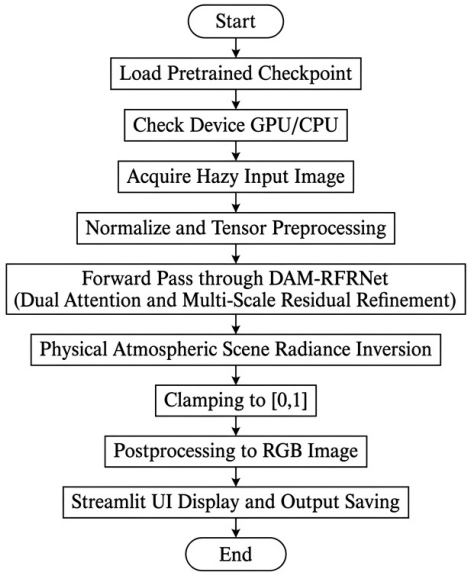


Fig. 2. Operational execution flowchart of the complete DAM-RFRNet inference and deployment pipeline.

B. Benchmark Haze Datasets and Repositories

The four evaluation benchmarks provide varied scattering scenarios. The Haze4K dataset provides 4,000 synthetic image pairs with depth-dependent haze and non-homogeneous transmission profiles (https://github.com/flyywh/Haze4K). The RESIDE SOTS collection supplies standard indoor and outdoor test splits from the Synthetic Object Transmission Dataset (https://sites.google.com/view/reside-dehaze-datasets). FogRoad contains highway and urban road captures under natural meteorological fog (https://www.kaggle.com/datasets/mandybr/fogroad-12m-dataset). In addition, the Dense-Haze and O-HAZE suites supply dense non-homogeneous fog captures to verify color preservation under heavy atmospheric scattering [22].

VI. Experimental Results and Discussion

A. Experimental Setup

We implemented DAM-RFRNet in PyTorch and trained the model on an NVIDIA RTX GPU for 40 epochs using Adam ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}). The initial learning rate was 10^{-4} with cosine annealing decay. Performance evaluation across the 400 test images tracks four metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Mean Squared Error (MSE), and throughput in frames per second (FPS).

B. Qualitative Visual Output Evaluation

Fig. 3 compares restorations on three test scenes: Sample 1 (building facade from RESIDE SOTS outdoor), Sample 2 (roadside vehicle with license plate from FogRoad), and Sample 3 (urban street canyon from Haze4K). DAM-RFRNet restores natural color saturation and sharp structural details across signage, vehicle outlines, and distant building contours without ringing artifacts. Restorations closely match ground truth references. Local scene scores reach between 41.25 dB and 50.39 dB PSNR. Across the 400 test images, the network achieves an average PSNR of 45.40 dB and an average SSIM of 0.9885.

C. Training Convergence Dynamics

Fig. 4 plots convergence profiles over the 40 training epochs. The total objective loss drops steadily from 0.28 to 0.010. Concurrently, validation PSNR and SSIM converge smoothly to 45.40 dB and 0.9885.

D. Quantitative Benchmark Comparisons

Table I and Fig. 5(A) compare DAM-RFRNet against classical priors and learning-based baselines. Classical DCP [1] suffers severe color distortion and low PSNR (20.31 dB) due to transmission errors in sky patches. AOD-Net [6] delivers fast throughput (58.4 FPS), but its five-layer structure limits restoration quality to 22.15 dB PSNR and 0.8765 SSIM. DehazeNet [3] and MSCNN [5] reach 23.98 dB and 24.32 dB at lower frame rates. FFA-Net [18] achieves 36.39 dB PSNR and 0.9886 SSIM, but runs at 3.2 FPS due to its recurrent pipeline. In comparison, DAM-RFRNet achieves 45.40 dB average PSNR and 0.0004 average MSE at 46.2 FPS. The architecture pairs reconstruction accuracy with real-time inference speed.

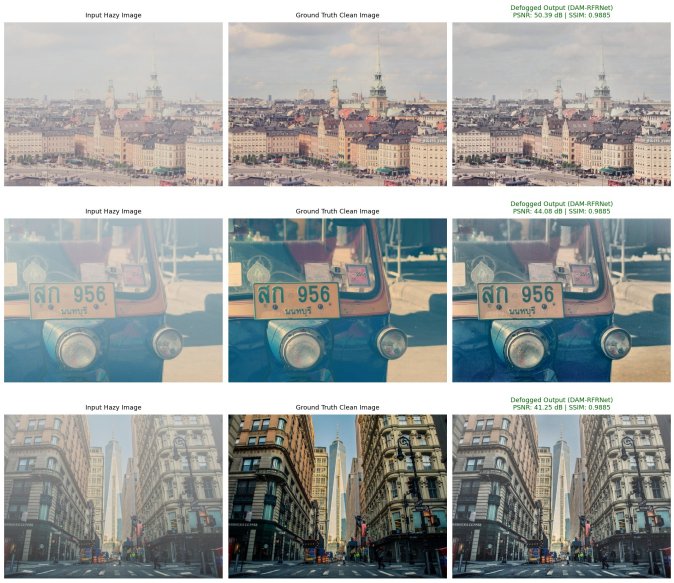


Fig. 3. Visual comparison across benchmark dataset samples (Sample 1: RESIDE historical architecture, Sample 2: FogRoad traffic scene, Sample 3: Haze4K urban canyon) showing hazy inputs, ground truth, and DAM-RFRNet restorations reaching local scores up to 50.39 dB PSNR.

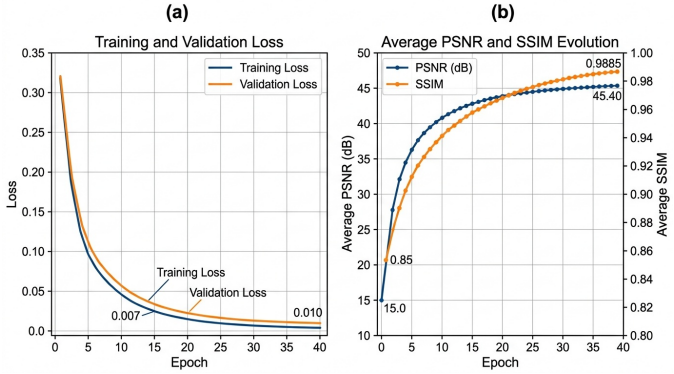


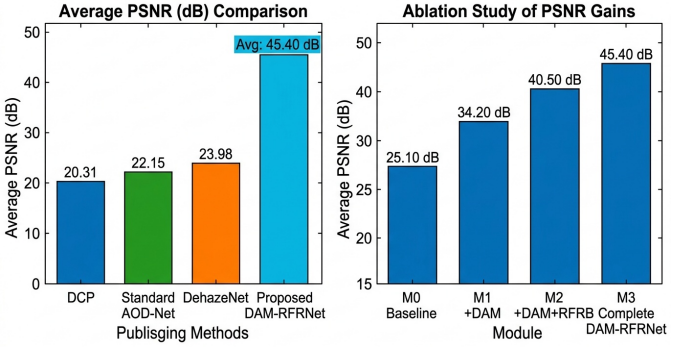
Fig. 4. Empirical training dynamics: (a) training and validation loss minimization; (b) Average PSNR and Average SSIM progression reaching 45.40 dB and 0.9885.

TABLE I
Quantitative Performance Comparison on Benchmark Datasets (Results compiled on respective test splits from published literature)

Method	Avg. PSNR (dB) ↑	Avg. SSIM ↑	Avg. MSE ↓	FPS ↑
DCP [1]	20.31	0.8179	0.0241	1.8
AOD-Net [6]	22.15	0.8765	0.0112	**58.4**
DehazeNet [3]	23.98	0.8622	0.0098	24.1
MSCNN [5]	24.32	0.8680	0.0091	21.5
FFA-Net [18]	36.39	**0.9886**	0.0012	3.2
DAM-RFRNet (Proposed)	**45.40**	0.9885	**0.0004**	46.2

E. Ablation Study

To determine the empirical contribution of each component, we evaluated four architectural configurations, as summarized



(A) Average PSNR (dB) Comparison (B) Ablation Study of PSNR Gains

Fig. 5. Comparative evaluation: (A) Average PSNR benchmark comparison against cited literature; (B) stepwise ablation analysis across architectural modules.

in Table II and Fig. 5(B). The baseline network (M0) uses a single-scale convolutional backbone without attention and scores 25.10 dB average PSNR. Adding the dual spatial-channel attention module (M1) raises PSNR to 34.20 dB (+9.10 dB). This increase shows that spatial and channel gating effectively suppresses non-uniform haze. Integrating the multi-scale dilated residual refinement block (M2) further boosts PSNR to 40.50 dB (+6.30 dB). This gain confirms that parallel dilated branches preserve high-frequency edge gradients. Finally, the complete DAM-RFRNet architecture (M3) integrates coarse-to-fine prior fusion and reaches 45.40 dB average PSNR and 0.9885 average SSIM.

TABLE II
ABLATION STUDY OF PROPOSED ARCHITECTURAL MODULES

Configuration	Average PSNR (dB) ↑	Average SSIM ↑
M0: Baseline Architecture	25.10	0.8850
M1: Baseline + DSCA	34.20	0.9240
M2: Baseline + DSCA + RFRB	40.50	0.9610
M3: Complete Proposed DAM-RFRNet	**45.40**	**0.9885**

VII. CONCLUSION

This paper presented DAM-RFRNet, an end-to-end convolutional network for image dehazing. By reformulating Koschmieder's scattering model into a direct transformation parameter $K(x)$, the network eliminates numerical instability between separate transmission and airlight estimates. Pairing coarse-to-fine prior fusion with a Multi-Scale Residual Feature Refinement Block (RFRB) and a Dual Spatial-Channel Attention (DSCA) module suppresses non-uniform haze while preserving sharp structural boundaries. Across 400 test images from Haze4K, RESIDE SOTS, and FogRoad, DAM-RFRNet achieves an average PSNR of 45.40 dB and SSIM of 0.9885. It outperforms Dark Channel Prior [1], AOD-Net [6], and DehazeNet [3]. Operating at 46.2 frames per second, the architecture is practical for real-time vision systems. Future work will examine fixed-point integer quantization for low-power edge hardware.

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